ZEROS OF THE INDEPENDENCE POLYNOMIALS OF UNIFORM LINEAR HYPERCOMBS

KNOWLEDGE SENSE

Abstract. For $n \geq 1$ and $\ell \geq 3$ let $C_{n,\ell}$ be the ℓ-uniform linear hypercomb whose spine has n edges. We prove that every zero z of its strong independence polynomial satisfies $|\arg z| > 2\pi/3$. By a classical criterion of Brenti, Royle and Wagner, the coefficients of this polynomial are therefore strictly log-concave, and in particular unimodal. This answers, for all n and ℓ, a question of Galvin and Sharpe, who proved unimodality outside a window of length $O(\sqrt{n})$. Up to a linear factor, the polynomial is the determinant of a tridiagonal matrix with two corner corrections. Testing a unit kernel vector against this matrix reduces the problem to a three-parameter family of quartic polynomials, all of whose zeros we locate.

1. Introduction

Let $\mathcal{H}$ be a hypergraph. A set of vertices of $\mathcal{H}$ is a *strong independent set* if it meets every edge of $\mathcal{H}$ in at most one vertex. If $i_k(\mathcal{H})$ denotes the number of strong independent sets of size k, we write

$$I(\mathcal{H}; x) = \sum_{k \geq 0} i_k(\mathcal{H})\, x^k$$

for the (strong) independence polynomial. For graphs this is the usual independence polynomial, and whether every tree has a unimodal independence sequence is a well-known open question of Alavi, Malde, Schwenk and Erdős [1]. Galvin and Sharpe [4] began the study of the corresponding question for hypertrees. They restricted attention to *linear* hypertrees, in which two edges share at most one vertex, and showed that linear hyperpaths have real-rooted, and linear hyperstars log-concave, strong independence polynomials [4, Theorem 1.3].

The next family considered in [4] consists of hypercombs. Let $n \geq 0$ and $\ell \geq 2$. The *ℓ-uniform linear hypercomb* $C_{n,\ell}$ has spine vertices $u_0, \ldots, u_n$ and $2n+1$ edges of size ℓ: for $1 \leq i \leq n$ a *spine edge* consisting of u_{i-1}, u_i and $\ell - 2$ further vertices, and for $0 \leq i \leq n$ a *tooth* consisting of u_i and $\ell - 1$ further vertices. All vertices other than $u_0, \ldots, u_n$ lie in exactly one edge. For $n \geq 1$ this is the hypercomb of [4, Section 1.1]; for $n = 1$ the two teeth are attached at distinct vertices of the unique spine edge, as required there. The hypergraph $C_{0,\ell}$ is a single edge, and $I(C_{0,\ell}; x) = 1 + \ell x$ is the initial condition used in [4, Theorem 1.7]. For $\ell = 2$ the hypercomb is the centipede, whose independence polynomial is real-rooted [7].

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For $\ell \geq 3$ the situation is different. Galvin and Sharpe found a linear recurrence for $I(C_{n,\ell}; x)$ [4, Theorem 1.7] and observed by computer that its coefficient sequence is unimodal for $n \leq 500$ and $\ell \leq 12$. They proved that for every $\ell \geq 3$ there are constants $K_\ell, L_\ell > 0$ such that the sequence is weakly increasing up to the index $K_\ell n - L_\ell \sqrt{n}$ and weakly decreasing from the index $K_\ell n + L_\ell \sqrt{n}$ [4, Corollary 1.8]; an earlier claim of log-concavity for all n and ℓ had been withdrawn [4, Note 1.9]. They asked whether the window of length $O(\sqrt{n})$ can be shortened or eliminated [4, Question 1.10(4)]. We eliminate it by locating the zeros. Throughout, arg denotes the principal argument, with values in $(-\pi, \pi]$.

Theorem 1.1. *Let $n \geq 1$ and $\ell \geq 3$. Every zero z of $I(C_{n,\ell}; x)$ satisfies $|\arg z| > 2\pi/3$.*

Since $I(C_{n,\ell}; 0) = 1$, the theorem places all zeros in the open sector of opening $2\pi/3$ around the negative real axis. A classical sufficient condition of Brenti, Royle and Wagner [2, Proposition 1.2], in the form recorded by Brown and Cameron [3, Corollary 2.3] and restated as Proposition 4.2 below, converts this into information on the coefficients.

Corollary 1.2. *Let $n \geq 1$ and $\ell \geq 3$, and write $I(C_{n,\ell}; x) = \sum_{k=0}^{2n+1} c_k x^k$. Then $c_k > 0$ for $0 \leq k \leq 2n+1$, and*

$$c_k^2 > c_{k-1} c_{k+1} \qquad (1 \leq k \leq 2n).$$

In particular the sequence $c_0, c_1, \ldots, c_{2n+1}$ is log-concave and unimodal.

Thus the window in [4, Corollary 1.8] is empty for every $n \geq 1$ and $\ell \geq 3$, which settles [4, Question 1.10(4)]. The parts of [4, Question 1.10] concerning all linear hypertrees, or all uniform linear hypertrees, remain open. We emphasize that the new statement is Theorem 1.1; the passage from the location of zeros to Corollary 1.2 uses only the classical criterion.

A recent preprint [6] treats a weighted family of hypercombs, defined by the analogous recurrence with spine and tooth weights $s, t > 0$; the uniform hypercomb $C_{n,\ell}$ corresponds to $s = \ell - 2$ and $t = \ell - 1$. It is shown there that for each fixed pair $s, t > 0$ there is $N(s,t)$ such that the coefficients are strictly log-concave for every $n \geq N(s,t)$, and the question for all lengths in the unweighted case is left open. Corollary 1.2 applies only to the uniform weights, but covers every length $n \geq 1$, and Theorem 1.1 gives in addition the location of the zeros. To our knowledge, Theorem 1.1 has not appeared in the literature.

Remark 1.3. The sector cannot be replaced by the negative real axis. For $n = 5$ and $\ell = 3$ the coefficients $c_0, \ldots, c_{11}$ of $I(C_{5,3}; x)$ are

$$1,\ 23,\ 220,\ 1156,\ 3716,\ 7700,\ 10564,\ 9716,\ 5984,\ 2400,\ 576,\ 64,$$

as one checks from the recurrence (2.2) below. Here $c_{10}^2 = 331776 < 337920 = \frac{11}{10} \cdot 2 \cdot c_9 c_{11}$, so Newton's inequality $c_k^2 \geq (1 + 1/k)(1 + 1/(d-k)) c_{k-1} c_{k+1}$, valid for every polynomial of degree d with positive coefficients and only real zeros, fails for $d = 11$ and $k = 10$. Hence $I(C_{5,3}; x)$ has non-real zeros, in contrast with the hyperpaths of [4, Theorem 1.3].

Method. Up to the factor $1 + (\ell - 1)x$, the polynomial $I(C_{n,\ell}; x)$ is the determinant of an $n \times n$ tridiagonal matrix whose corner entries are perturbed by x (Lemmas 2.2 and 2.3). At a zero, a unit kernel vector of this matrix produces two real numbers

σ and t, the values of the quadratic forms of the corner and path matrices, with $0 \le \sigma \le 1$ and $|t| \le 2$, and z is a zero of an explicit quartic depending on σ and $|t|$ (Lemma 2.4). The heart of the proof is Lemma 3.1, which locates the zeros of all these quartics. After the substitutions $w = -1/z$ and $w = a + v^2$, with $a = \ell - 1$, it becomes a statement about a quartic in v with exactly two real zeros, and Vieta's formulas determine the position of the remaining pair. No realizability of the parameters is needed: the lemma covers the whole range $0 \le \sigma \le 1$ and $|t| \le 2$. The argument works verbatim with a real parameter $a \ge 2$ in place of $\ell - 1$ (Theorem 4.1).

Dependencies. The proof of Theorem 1.1 is self-contained. It uses the recurrence of [4, Theorem 1.7], which we rederive in Lemma 2.1 to fix conventions. Corollary 1.2 uses in addition the criterion of [2].

Organization. Section 2 derives the recurrence and the determinant formula and reduces Theorem 1.1 to a family of quartics. Section 3 locates the zeros of these quartics. Section 4 completes the proofs.

2. A determinant formula

Let $a \ge 1$ be real, and put

$$f = 1 + ax, \qquad g = 1 + (a-1)x, \qquad A = fg, \qquad B = xf. \tag{2.1}$$

Define polynomials $C_n = C_n(x)$ by

$$C_0 = f + x, \qquad C_1 = f(A + 2x), \qquad C_n = AC_{n-1} + BC_{n-2} \quad (n \ge 2). \tag{2.2}$$

Since $A(0) = 1$ and $B(0) = 0$, we have $C_n(0) = 1$ for all $n \ge 0$.

Lemma 2.1. *Let $\ell \ge 2$ and $a = \ell - 1$. Then $I(C_{n,\ell}; x) = C_n(x)$ for every $n \ge 0$.*

Proof. Let U_n and V_n be the generating polynomials, by size, of the strong independent sets of $C_{n,\ell}$ that do not contain, respectively contain, the last spine vertex u_n. The single tooth of $C_{0,\ell}$ gives $U_0 = 1 + (\ell - 1)x = f$ and $V_0 = x$.

The hypercomb $C_{n+1,\ell}$ arises from $C_{n,\ell}$ by adding a spine edge $\{u_n, u_{n+1}\} \cup X$ with $|X| = \ell - 2$ and a tooth $\{u_{n+1}\} \cup Y$ with $|Y| = \ell - 1$, where X, Y and $\{u_{n+1}\}$ consist of new vertices. A set S of vertices of $C_{n+1,\ell}$ is a strong independent set if and only if its restriction S' to $C_{n,\ell}$ is one and the following holds. If $u_{n+1} \notin S$, then S contains at most one vertex of Y, and at most one vertex of X, none if $u_n \in S'$. If $u_{n+1} \in S$, then $u_n \notin S'$ and S contains no vertex of $X \cup Y$. Since $1 + (\ell - 2)x = g$, this gives

$$U_{n+1} = f\,(gU_n + V_n), \qquad V_{n+1} = xU_n.$$

Thus

$$\begin{pmatrix} U_n \\ V_n \end{pmatrix} = M^n \begin{pmatrix} U_0 \\ V_0 \end{pmatrix}, \qquad M = \begin{pmatrix} A & f \\ x & 0 \end{pmatrix}.$$

The matrix M has trace A and determinant $-B$, so $M^2 = AM + BI$ by the Cayley–Hamilton theorem, and $U_n + V_n$ satisfies the recurrence in (2.2) for $n \ge 2$. Finally $U_0 + V_0 = f + x$ and $U_1 + V_1 = f(fg + x) + xf = f(A + 2x)$. □

Since $A = fg$ and $B = xf$, the recurrence (2.2) is $C_n = f\left(gC_{n-1} + xC_{n-2}\right)$, which is the recurrence of [4, Theorem 1.7]; one also checks that $C_1 = 1 + (3a + 1)x + 3a^2x^2 + a^2(a-1)x^3$ agrees with the initial condition there.

Define $F_{-1} = 0$, $F_0 = 1$ and $F_j = AF_{j-1} + BF_{j-2}$ for $j \ge 1$, so that $F_1 = A$.

Lemma 2.2. *For every $n \geq 1$,*

$$C_n = f\left(F_n + 2xF_{n-1} + x^2F_{n-2}\right). \tag{2.3}$$

Proof. Denote the bracket by R_n. Then $R_1 = A + 2x$, so (2.3) holds for $n = 1$. For $n = 2$,

$$C_2 = AC_1 + BC_0 = f(A^2 + 2xA) + xf(f + x) = f(A^2 + B + 2xA + x^2) = fR_2.$$

For $n \geq 3$ the indices n, $n-1$ and $n-2$ are positive, so each of F_n, F_{n-1}, F_{n-2} satisfies the defining recurrence and $R_n = AR_{n-1} + BR_{n-2}$. Since fR_n and C_n satisfy the same recurrence and agree for $n = 1, 2$, they agree for all $n \geq 1$. □

For $n \geq 2$ let J_n be the adjacency matrix of the path with n vertices, that is, the $n \times n$ matrix with ones on the first super- and subdiagonal and zeros elsewhere, and let E_n be the diagonal matrix with diagonal $(1, 0, \ldots, 0, 1)$.

Lemma 2.3. *Let $n \geq 2$, $z \in \mathbb{C}$, and choose $\beta \in \mathbb{C}$ with $\beta^2 = B(z)$. Then the matrix*

$$M_n(z) = A(z)\, I_n + z\, E_n + i\beta\, J_n$$

satisfies $\det M_n(z) = F_n(z) + 2zF_{n-1}(z) + z^2F_{n-2}(z)$.

Proof. Let T_m be the $m \times m$ tridiagonal matrix with diagonal entries $A(z)$ and off-diagonal entries $i\beta$. Expansion along the last row gives $\det T_m = A(z) \det T_{m-1} - (i\beta)^2 \det T_{m-2} = A(z) \det T_{m-1} + B(z) \det T_{m-2}$, with $\det T_0 = 1$ and $\det T_1 = A(z)$; hence $\det T_m = F_m(z)$. The matrix $M_n(z)$ arises from T_n by adding z to the entries $(1, 1)$ and (n, n), which lie in distinct rows because $n \geq 2$. By multilinearity of the determinant in these two rows,

$$\det M_n(z) = \det T_n + z \det T_{n-1} + z \det T_{n-1} + z^2 \det T_{n-2},$$

where the three minors are obtained by deleting row and column 1, row and column n, and both; for $n = 2$ the last one is the empty matrix, with determinant 1. This is the asserted formula. □

Lemma 2.4. *Let $n \geq 2$ and let $z \in \mathbb{C}$ satisfy $C_n(z) = 0$. Then $z \neq 0$, and either $z = -1/a$ or there are real numbers $\sigma \in [0, 1]$ and $\tau \in [0, 2]$ such that*

$$\big(A(z) + \sigma z\big)^2 + \tau^2 z f(z) = 0. \tag{2.4}$$

Proof. We have $z \neq 0$ because $C_n(0) = 1$. Suppose $z \neq -1/a$, so that $f(z) \neq 0$. By Lemmas 2.2 and 2.3, $\det M_n(z) = 0$. Let $v \in \mathbb{C}^n$ satisfy $M_n(z)v = 0$ and $v^*v = 1$, and put

$$\sigma = v^*E_nv = |v_1|^2 + |v_n|^2, \qquad t = v^*J_nv = 2\,\mathrm{Re} \sum_{j=1}^{n-1} \overline{v_j}\, v_{j+1}.$$

Then $\sigma \in [0, 1]$, t is real, and $|t| \leq \sum_{j=1}^{n-1} \big(|v_j|^2 + |v_{j+1}|^2\big) \leq 2$. Moreover

$$0 = v^*M_n(z)v = A(z) + \sigma z + i\beta t.$$

Hence $(A(z) + \sigma z)^2 = -\beta^2 t^2 = -t^2 z f(z)$, and (2.4) holds with $\tau = |t|$. □

3. A family of quartics

Lemma 3.1. *Let $a \geq 2$, $\sigma \in [0,1]$ and $\tau \in [0,2]$ be real. Every zero z of*

$$\Phi(z) = \big((1+az)(1+(a-1)z) + \sigma z\big)^2 + \tau^2 z(1+az)$$

satisfies $|\arg z| > 2\pi/3$.

Proof. If $\tau = 0$, then Φ is the square of $1 + (2a-1+\sigma)z + a(a-1)z^2$. This quadratic has positive coefficients and discriminant $(2a-1+\sigma)^2 - 4a(a-1) = 1 + 2\sigma(2a-1) + \sigma^2 > 0$, so its zeros are negative real numbers. We assume from now on that $\tau > 0$.

Step 1: two substitutions. Since $\Phi(0) = 1$, every zero z of Φ is nonzero, and we put $w = -1/z$. From $w(1+az) = w - a$, $w(1+(a-1)z) = w - a + 1$ and $wz = -1$ we obtain

$$w^4\,\Phi(-1/w) = D(w)^2 - \tau^2 w^2 (w-a), \qquad D(w) = (w-a)(w-a+1) - \sigma w.$$

If $|\arg w| < \pi/3$, then $|\arg z| = \pi - |\arg w| > 2\pi/3$. It therefore suffices to show that every zero w of $D(w)^2 - \tau^2 w^2(w-a)$ satisfies $|\arg w| < \pi/3$. Let w be such a zero, and choose $v \in \mathbb{C}$ with $v^2 = w - a$. Then $D(w)^2 = (\tau w v)^2$, and after replacing v by $-v$ if necessary, $D(w) = \tau w v$. Substituting $w = a + v^2$, so that $D(w) = v^2(v^2+1) - \sigma(a+v^2)$, we find that v is a zero of

$$\Psi(v) = v^4 - \tau v^3 + (1-\sigma)v^2 - a\tau v - a\sigma.$$

Consequently it suffices to prove that

$$|\arg(a+v^2)| < \pi/3 \qquad \text{for every zero } v \text{ of } \Psi. \tag{3.1}$$

Step 2: the real zeros of Ψ. For $0 < v \leq \tau$ we have $v^3(v-\tau) \leq 0$ and $(1-\sigma)v^2 \leq (1-\sigma)\tau v$, hence

$$\Psi(v) \leq (1-\sigma-a)\tau v - a\sigma < 0,$$

because $a > 1 \geq 1 - \sigma$. For $v \geq \tau$ we have $\Psi''(v) = 6v(2v-\tau) + 2(1-\sigma) > 0$. As $\Psi(\tau) < 0$ and $\Psi(v) \to \infty$, the strictly convex function Ψ has exactly one zero r in (τ, ∞), and $\Psi'(r) > 0$. So $r > \tau$ is the only positive zero of Ψ, it is simple, and $\Psi < 0$ on $(0, r)$. For $p \geq 0$,

$$\Psi(-p) = p^4 + \tau p^3 + (1-\sigma)p^2 + a\tau p - a\sigma$$

is strictly increasing in p, with derivative at least $a\tau > 0$, starts at $-a\sigma \leq 0$ and tends to infinity. Hence Ψ has exactly one zero $-p$ with $p \geq 0$, and it is simple. Since Ψ has real coefficients, its remaining zeros form a pair $\xi, \overline{\xi}$ with $\operatorname{Im}\xi \neq 0$. Moreover $p < r$: this is clear if $p = 0$, and if $p > 0$ then $\Psi(p) = \Psi(-p) - 2\tau p(p^2 + a) < 0$, so $p \in (0, r)$.

For the real zeros, $a + r^2$ and $a + p^2$ are positive, so (3.1) holds. The zero $\overline{\xi}$ yields the complex conjugate of $a + \xi^2$, so it remains to show that $|\arg(a+\xi^2)| < \pi/3$.

Step 3: Vieta's formulas. Put

$$\delta = r - p - \tau, \qquad Q = |\xi|^2 > 0, \qquad H = rp \geq 0.$$

Comparing the coefficients of v^3 in $\Psi(v) = (v-r)(v+p)(v-\xi)(v-\overline{\xi})$ gives $r - p + 2\operatorname{Re}\xi = \tau$, that is, $2\operatorname{Re}\xi = -\delta$. Hence $(v-\xi)(v-\overline{\xi}) = v^2 + \delta v + Q$, while

$(v-r)(v+p) = v^2 - (\tau+\delta)v - H$. Comparing the remaining coefficients, we obtain

$$Q = 1 - \sigma + H + \delta(\tau + \delta), \tag{3.2}$$

$$a = Q + \delta\,\frac{Q+H}{\tau}, \tag{3.3}$$

$$HQ = a\sigma. \tag{3.4}$$

Step 4: $\delta > 0$. Suppose that $\delta \le 0$. Since $r > p$, we have $\delta > -\tau$, and therefore $\delta(\tau+\delta) \le 0$. By (3.3), $a \le Q$, so (3.4) gives $H = a\sigma/Q \le \sigma$. Then (3.2) gives $Q \le 1 - \sigma + \sigma = 1$, which contradicts $Q \ge a \ge 2$.

Step 5: the sector. Put $K = (Q+H)/\tau$. Then $K > 0$, $K \ge Q/2$ because $H \ge 0$ and $\tau \le 2$, and $a = Q + \delta K$ by (3.3). Write $\xi = -\delta/2 + iy$ with $y \ne 0$, so that $Q = \delta^2/4 + y^2$. Then

$$a + \xi^2 = Q + \delta K + \frac{\delta^2}{4} - y^2 - i\delta y = \delta\Big(K + \frac{\delta}{2}\Big) - i\delta y.$$

The real part is positive by Step 4. Hence $|\arg(a+\xi^2)| < \pi/3$ if and only if $\delta^2 y^2 < 3\delta^2(K + \delta/2)^2$, that is,

$$Q < 3K^2 + 3K\delta + \delta^2. \tag{3.5}$$

If $Q \ge 4/3$, then $3K^2 \ge 3Q^2/4 \ge Q$, and (3.5) holds because $3K\delta + \delta^2 > 0$. If $Q < 4/3$, then $3K\delta = 3(a-Q) > 3(2 - 4/3) = 2 > Q$, and (3.5) holds again. This proves (3.1) and the lemma. □

4. Proofs of the main results

Theorem 4.1. *Let $a \ge 2$ be real and let C_n be defined by* (2.1) *and* (2.2). *For every $n \ge 0$, every zero z of C_n satisfies $|\arg z| > 2\pi/3$.*

Proof. The only zero of $C_0 = 1 + (a+1)x$ is $-1/(a+1)$. For $n = 1$, $C_1 = f\,(A + 2x)$, where f has the zero $-1/a$ and $A + 2x = 1 + (2a+1)x + a(a-1)x^2$ has positive coefficients and discriminant $8a+1 > 0$; so all zeros of C_1 are negative real numbers. Let $n \ge 2$ and let z be a zero of C_n. By Lemma 2.4, either $z = -1/a$, or (2.4) holds for some $\sigma \in [0,1]$ and $\tau \in [0,2]$. In the latter case z is a zero of the polynomial Φ of Lemma 3.1, because $A(z) = (1+az)(1+(a-1)z)$ and $zf(z) = z(1+az)$; hence $|\arg z| > 2\pi/3$. □

Proof of Theorem 1.1. Apply Lemma 2.1 and Theorem 4.1 with $a = \ell - 1 \ge 2$. □

The passage to coefficients rests on the following classical criterion. Brenti, Royle and Wagner [2, Proposition 1.2] state it for zeros in the sector $|\arg z| < \pi/3$; replacing $p(x)$ by $p(-x)$ changes the sign of every second coefficient and does not affect the inequalities, which gives the form below [3, Corollary 2.3].

Proposition 4.2 ([2, 3])**.** *Let $p(x) = \sum_{k=0}^{d} b_k x^k \in \mathbb{R}[x]$ have degree d and $p(0) \ne 0$, and suppose that every zero z of p satisfies $|\arg z| > 2\pi/3$. Then the coefficients $b_0, \dots, b_d$ are all positive or all negative, and $b_k^2 > b_{k-1}b_{k+1}$ for $1 \le k \le d-1$.*

Proof of Corollary 1.2. The $2n+1$ edges of $C_{n,\ell}$ cover all its vertices, so a strong independent set has at most $2n+1$ elements. Choosing one vertex outside the spine in every edge, which is possible because $\ell \ge 3$, gives one of size $2n+1$. Hence $I(C_{n,\ell}; x)$ has degree $2n+1$. By Theorem 1.1 and Proposition 4.2, its coefficients have a common sign, which is positive because $c_0 = 1$, and $c_k^2 > c_{k-1}c_{k+1}$ for

$1 \le k \le 2n$. Therefore the ratios c_k/c_{k-1}, $1 \le k \le 2n+1$, are strictly decreasing, and the sequence is unimodal. □

Remark 4.3. For the non-strict inequalities $c_k^2 \ge c_{k-1}c_{k+1}$ one can also argue directly. By Theorem 1.1, $I(C_{n,\ell}; x)$ is a product of factors $1 + x/\rho$ and $1 - (2\cos\theta)x/\rho + x^2/\rho^2$ with $\rho > 0$ and $2\pi/3 < |\theta| < \pi$. Since $-2\cos\theta \in (1, 2)$, every factor has positive and log-concave coefficients, and the convolution of positive log-concave sequences without internal zeros is again of this kind; see, for example, [5].

Remark 4.4. By Theorem 4.1 and Proposition 4.2, the conclusion of Corollary 1.2 holds for the polynomials C_n, $n \ge 1$, for every real $a \ge 2$. The assumption $a \ge 2$ is used in Steps 4 and 5 of the proof of Lemma 3.1. The case $a = 1$ corresponds to the centipede, whose independence polynomial is real-rooted [7].

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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